

EDUCATION

B.S. Computer Science · The University of Texas at Dallas

Expected Fall 2027

PROJECTS

Event-Driven Commerce Platform

v1.0.0 · Aug 2026

Java 21 · Spring Boot · Kafka · PostgreSQL · Flyway · Docker · Testcontainers · GitHub Actions

- Built three independently deployable services (order, inventory, payment) that coordinate an order → inventory → payment saga over Kafka with no synchronous calls and no shared database.
- Implemented a transactional outbox in every service with a multi-replica-safe publisher (FOR UPDATE SKIP LOCKED, bounded 8 s lock hold, cross-replica per-order ordering guard).
- Made every consumer idempotent with a processed-events ledger claimed inside the mutation's transaction; classified failures into business outcomes, contract violations (dead-letter), and transient errors (exponential backoff).
- Added optimistic locking on the order aggregate and inventory rows, saga compensation that releases held stock exactly once, and three-layer at-most-once payment charging.
- 345 automated tests against real PostgreSQL and Kafka, including latch-driven concurrency proofs and a 16-test end-to-end suite running all services as containers; all suites run in CI.

Case study: ahmet-keles.github.io/projects/event-driven-commerce · Source: github.com/ahmet-keles/event-driven-commerce

K-Map Looper

Live · Sep 2026

HTML · Canvas · Vanilla JavaScript

Single-file browser tool for digital-logic coursework: fill a 2/3/4-variable Karnaugh map, draw colored group loops with wrap-around, and export a clean PNG. Draws, doesn't solve. Live at ahmet-keles.github.io/kmap-looper.

NovaSearch · Distributed Job Scheduler · Distributed Object Storage

In development

Python · FastAPI · LangChain · pgvector · Redis | Java · Spring Boot · PostgreSQL · Docker

Hybrid keyword + vector search with a RAG-oriented architecture; a background-job platform focused on scheduling, leases, retries, and execution tracking; and an object store exploring chunking, replication, checksums, and node-failure handling.

EXPERIENCE

Superintendent Intern · Keles Group Inc.

Summer 2026

Supported superintendent teams on construction projects totaling over \$10.2M across Wylie and Grand Prairie, TX.

Customer Experience Intern · DTF Order

Summer 2024

Supported customer-facing operations for a printing business in Richardson, TX.

SKILLS

Languages	Java, Python, C++, SQL, JavaScript, C#
Backend	Spring Boot, FastAPI, REST APIs, gRPC
Data & messaging	PostgreSQL, Kafka, Redis, RabbitMQ, pgvector
Infrastructure	Docker, Git, Linux, CI/CD (GitHub Actions), AWS, Azure, Testcontainers

CERTIFICATIONS

Claude 101 · Anthropic

Aug 2026